

Pose-Based Human Activity Classifier

Computer Vision - Complex Computing Problem (CLO-3)

Video source: own pre-recorded clip 'activity.mp4' (~20.5 s, 613 frames @ 30 fps); a single person alternating between standing and sitting.

1. Angle Computation Approach

Body keypoints were extracted from every frame with the pre-trained MediaPipe Pose model (33 landmarks per frame). Missing detections were linearly interpolated and the keypoint trajectories were de-jittered with a Savitzky-Golay filter (window 11, order 3) along the time axis. For three landmarks A-B-C, the joint angle at B is the angle between vectors B->A and B->C, computed as $\arccos((\mathbf{BA} \cdot \mathbf{BC}) / (|\mathbf{BA}| |\mathbf{BC}|))$ in pixel coordinates. Three angles were tracked, averaging left and right sides: knee (hip-knee-ankle), hip (shoulder-hip-knee) and elbow (shoulder-elbow-wrist).

2. Rule-Based Logic

Standing shows an extended knee (~150-180 deg) and an open hip (~150-180 deg); sitting shows a bent knee (~65-95 deg) and a folded hip. The classifier labels a frame as STANDING if (knee angle > 130 deg) AND (hip angle > 130 deg), otherwise SITTING. Using two joints rather than the knee alone adds robustness to single-joint noise.

3. Accuracy Results

Overall accuracy: 99.35% (609/613 frames) against a manually-labeled per-frame ground truth.

Confusion matrix (rows=true, cols=pred):

	sitting	standing
sitting	429	0
standing	4	180

Mismatched frames: [140, 141, 416, 417]

4. Observations

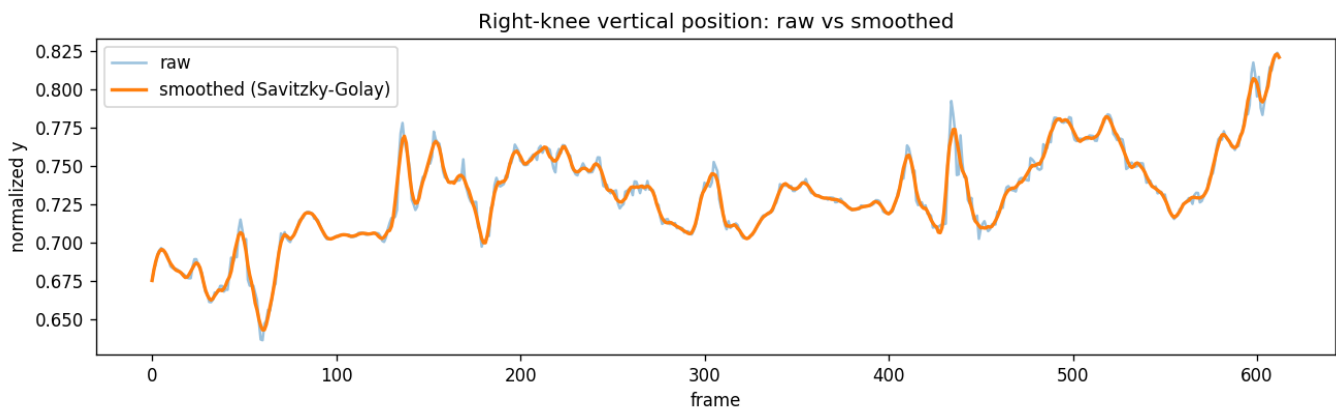
- MediaPipe detected the subject in all 613 frames; Savitzky-Golay smoothing removed jitter without distorting real motion.
- Knee and hip angles separate the two activities cleanly; the elbow angle is comparatively uninformative for stand/sit.
- All classification errors fall on transition frames (mid stand-up / sit-down), where the posture is genuinely ambiguous - the expected place for a hard threshold to disagree with a discrete human label.

Demo Screenshots

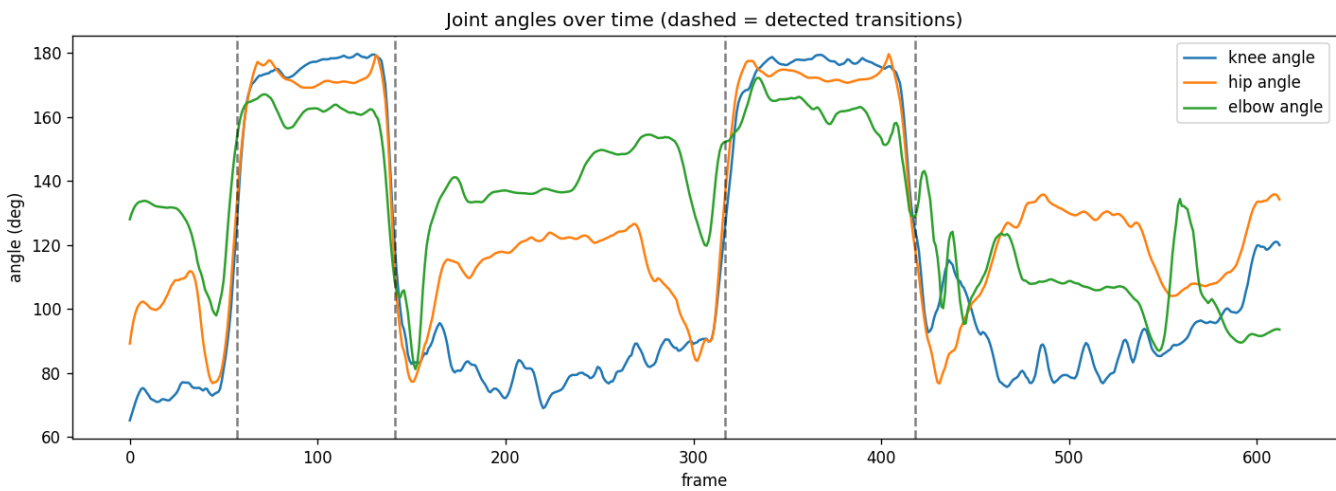


Skeleton overlay: standing (left) and sitting (right)

Raw vs smoothed keypoint trajectory



Joint angles over time (dashed = transitions)



Predicted vs ground truth

Predicted vs ground truth (accuracy 99.35%)

